

30-60K

5.2M



10:20
10:20
10:40
10:45
11:15
11:20
10:00

DELAYED
DELAYED
DELAYED
DELAYED
DELAYED
DELAYED

The goal:

Building a prediction model for the odds
of a flight being delayed



Step 1

Finding
Data

Requirements for our Data:

- **Reputable**
 - Reliable source
 - Minimal missing values
- **Completeness**
 - Data from 2015-2024
 - Exclude outlier years
- **Well Organized**
 - Easy to understand
 - Additional tables to cross reference
- **Diverse Variables**
 - Customizable variables

Bureau of Transportation Statistics

Ask-A-LibrarianA-Z Index

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Data Finder

By Mode

Aviation
Maritime
Highway
Transit
Rail
Pipeline
Bike/Pedestrian
Other

By Subject

Safety
Freight Transport
Passenger Travel
Infrastructure
Economic/Financial
Social/Demographic
Energy
Environment
National Security

On-Time : Reporting Carrier On-Time Performance (1987-present)

Latest Available Data: July 2025

DatabasesData TablesTable Contents

Filter GeographyFilter YearFilter Period

Download Instructions

All2025January

☐ Prezipped File☐ % Missing in table☐ Documentation☐ Term

Download

Field Name	Description	Support Table
Time Period		
[No Title]	Year	
☐ Quarter	Quarter (1-4)	Get Lookup Table
☐ Month	Month	Get Lookup Table
☐ DayofMonth	Day of Month	
☐ DayOfWeek	Day of Week	Get Lookup Table
☐ FlightDate	Flight Date (yyyymmdd)	
Airline		
☐ Reporting_Airline	Unique Carrier Code. When the same code has been used by multiple carriers, a numeric suffix is used for earlier users, for example, PA, PA(1), PA(2). Use this field for analysis across a range of years.	Get Lookup Table

Variables

- **Date** - Year, Month, Day, Day of Week
- **Carrier** - Airport Carrier Code
- **Airport** – Code, City
- **Departure/Arrival Times** - Scheduled & Actual
- **Delay Times** - Difference between
- **Cancellation/Diversion** - Identifier
- **Distance** – Miles of flight
- **Air Time** – Minutes in aircraft in the air
- **Reasons for Delay** - Code

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA
1	YEAR	MONTH	DAY_OF_MONTH	DAY_OF_WEEK	FL_DATE	OP_UNIQUE	OP_CARRIER	ORIGIN_AIRPORT	ORIGIN_CITY	DEST_AIRPORT	DEST_CITY	CRS_DEP_TIME	DEP_TIME	DEP_DELAY	CRS_ARR_TIME	ARR_TIME	ARR_DELAY	CANCELLATION	CANCELLATION	DIVERTED	AIR_TIME	FLIGHTS	DISTANCE	CARRIER_D	WEATHER		
2	2015	8	1	6	##### AA	19805	10140	ABQ	Albuquerque	11057	CLT	Charlotte, NC	40	34	-6	605	536	-29	0	0	163	1	1449				
3	2015	8	1	6	##### AA	19805	10140	ABQ	Albuquerque	11298	DFW	Dallas/Fort	623	618	-5	912	904	-8	0	0	78	1	569				
4	2015	8	1	6	##### AA	19805	10140	ABQ	Albuquerque	11298	DFW	Dallas/Fort	800	758	-2	1047	1042	-5	0	0	78	1	569				
5	2015	8	1	6	##### AA	19805	10140	ABQ	Albuquerque	11298	DFW	Dallas/Fort	1105	1059	-6	1350	1336	-14	0	0	80	1	569				
6	2015	8	1	6	##### AA	19805	10140	ABQ	Albuquerque	11298	DFW	Dallas/Fort	1209	1206	-3	1451	1439	-12	0	0	76	1	569				
7	2015	8	1	6	##### AA	19805	10140	ABQ	Albuquerque	11298	DFW	Dallas/Fort	1435	1432	-3	1721	1715	-6	0	0	76	1	569				
8	2015	8	1	6	##### AA	19805	10140	ABQ	Albuquerque	11298	DFW	Dallas/Fort	1619	1618	-1	1905	1906	1	0	0	79	1	569				
9	2015	8	1	6	##### AA	19805	10257	ALB	Albany, NY	11057	CLT	Charlotte, NC	800	759	-1	1016	956	-20	0	0	103	1	646				
10	2015	8	1	6	##### AA	19805	10257	ALB	Albany, NY	11057	CLT	Charlotte, NC	1435	1530	55	1650	1731	41	0	0	104	1	646	41	0		
11	2015	8	1	6	##### AA	19805	10299	ANC	Anchorage, AK	11298	DFW	Dallas/Fort	2040	2349	189	601	848	167	0	0	340	1	3043	0			

Step 2

Data
Wrangling

Dataset Sources

- Flight On-Time Performance Dataset
- U.S. Department of Transportation, Bureau of Transportation Statistics
- Used annual CSV files from **multiple years**
- Total size: **millions of records**
- Structured data with detailed flight information

Initial Challenges

- * Dataset was **too large to read comfortably**
- Raw files contained:
 - **Millions of rows**
 - **Multiple years**
 - **Different airports**
- **Solution:**
 - **We split the dataset into 11 Excel files**
 - Improved readability and local inspection

Filtering by Airports

To make the dataset specific to our project goals, we filtered for:

- **State College Airport (SCE)**
- **Nearby major hubs:**
 - Pittsburgh (PIT)
 - Philadelphia (PHL)
 - Newark (EWR)
 - New York (JFK/LGA)

Creating the Holiday Variable

- We engineered a flag feature: **is_holiday**
- TRUE if the flight date was between:
- **December 1 – January 10**
- This captures the **winter holiday travel spike**

Data Cleaning & Merging

Steps performed:

- Removed cancelled/diverted flights
- Selected only necessary columns
- Merged annual files into **one unified dataset**
- Validated missing values
- Prepared a separate validation dataset

Outcome of Preprocessing

- Dataset reduced to **clean and usable format**

- Features ready for modeling:

Departure time

Airline

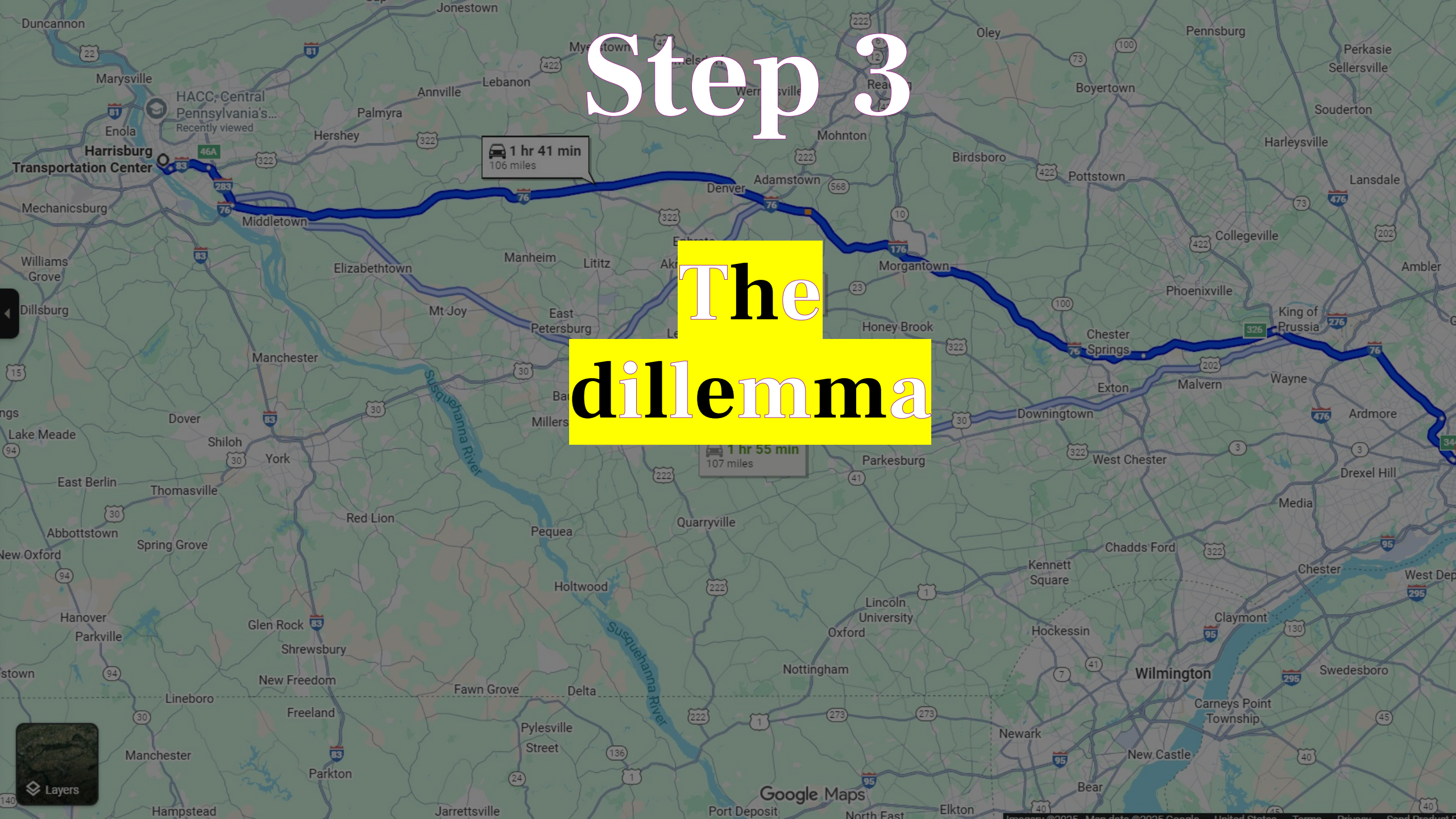
Origin/Destination

is_holiday

- Data was used directly for:
- **PySpark ingestion**
- **Feature engineering**
- **Model building**

Step 3

The dilemma





Random Forest

Scaling with ICDS Roar-Collab

- **1. Solving the 'Memory Wall' (RAM)**
 - **Local Problem:** Loading 14.5 million rows caused immediate RAM overflows/crashes.
 - **Roar-Collab Fix:** High-Memory Nodes (256GB+) allow us to load the *entire* raw dataset into memory at once. This enables full feature engineering without discarding potentially valuable data.
- **2. Accelerating XGBoost Training (CPU)**
 - **Local Problem:** Training complex trees on millions of rows takes hours on a standard laptop.
 - **Roar-Collab Fix:** XGBoost is designed for **Parallel Processing**. By running on Roar-Collab, we can utilize 8 CPU cores simultaneously reducing training time from hours to minutes.

- Each tree in forest provided random subset of overall data
- Each tree is grown to their max depth, and left as is (even if trees are overfit for certain data points)
- Final prediction takes the averages of the predictions made from each decision tree in the forest
 - These averages from many different trees help build a generalization of the dataset
- Random forest models are easy to understand, parameter tune, and draw variable importance

Regression

Metrics:

RMSE – Average error between actual and predicted delays

R² – How predictable a delay is (with variables given)

Classifier

Metrics:

F1 – mean of precision and recall

Precision – How many correct prediction made (across both classes)

Recall – How many true predictions made (across both classes)

Accuracy - Percentage of correct classifications

```
print("message: HPT FOR DELAY")
for numTrees in numTrees_list:
    print(f"message: trees: {numTrees}")
    for maxDepth in maxDepth_list:
        rf = RandomForestRegressor(
            featuresCol="features",
            labelCol="label",
            numTrees=numTrees,
            maxDepth=maxDepth,
            maxBins=256,
            seed=42
        )

        rf_classifier = RandomForestClassifier(
            featuresCol="features",
            labelCol="classifier_label",
            numTrees=numTrees,
            maxDepth=maxDepth,
            maxBins=256,
            seed=42
        )

        rf_model = rf.fit(train_set)
        rf_val_pred = rf_model.transform(val_set)
        rmse = evaluator.evaluate(rf_val_pred)

        if rmse < best_rmse:
            best_rmse = rmse
            best_reg_model = rf_model
            best_reg_config = (numTrees, maxDepth)

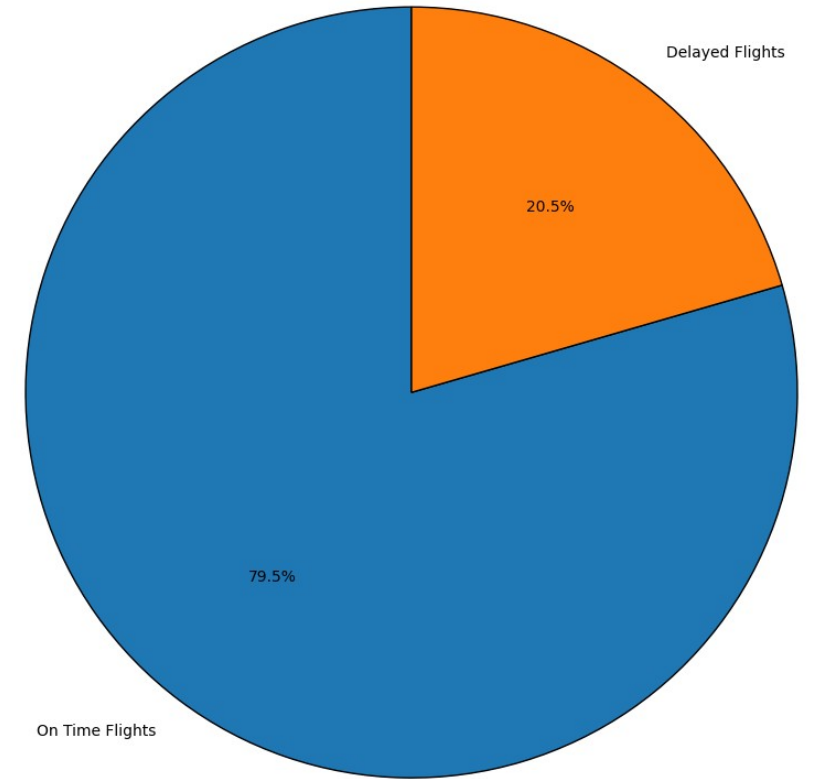
        rf_classifier_model = rf_classifier.fit(train_set)
        rf_classifier_val_pred = rf_classifier_model.transform(val_set)
        f1 = f1_classifier.evaluate(rf_classifier_val_pred)

        if f1 > best_f1:
            best_f1 = f1
            best_classifier_model = rf_classifier_model
            best_classifier_config = (numTrees, maxDepth)
```

•Problems Faced:

- Data of data severely affects random forest precision
- Classifier model is sensitive to majority class
- Delay threshold is a sensitive variable that is hard to define

Distribution of Delayed and On Time Flights



Why choose XGBoost?

- **Scalability:** Dataset contains **14.5 million rows**. Use XGBoost to run huge dataset without crashing
- **Handles Imbalance:** Our data is 80% On-Time vs. 20% Delayed. XGBoost has a built-in parameter (`scale_pos_weight`) that re-balances the data automatically during training.
- **Tabular Dominance:** For structured data (Excel-style rows and columns), XGBoost consistently outperforms deep learning or simple linear regression.

Code for Xgboost

```
model = xgb.XGBClassifier(  
    objective='binary:logistic',  
    eval_metric='logloss',  
    scale_pos_weight=2.5, # Tuned value  
    n_estimators=200,  
    max_depth=7,  
    learning_rate=0.1,  
    random_state=42,  
    n_jobs=-1  
)
```

Key Hyperparameters:

- **scale_pos_weight = 2.5**: Heavily penalizes the model for missing a delay. This effectively counteracts the 80/20 class imbalance.
- **max_depth = 7**: Increased to allow the model to learn more complex, specific patterns(Also to avoid(overfitting)
- **n_estimators = 200**: The number of "boosting rounds" (trees). We increased this to ensure the model has enough iterations to correct its errors.

Step 4

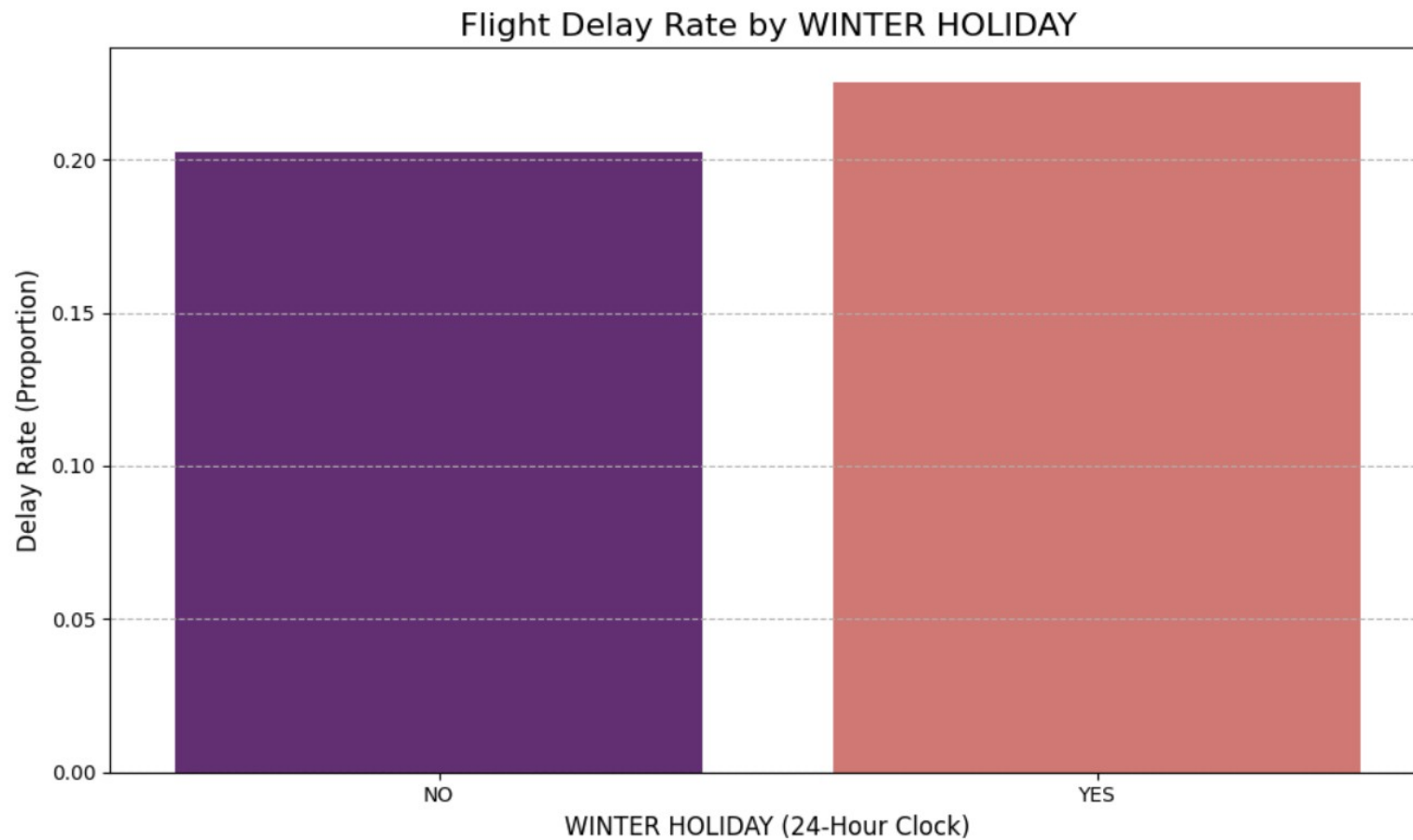
Visualization & Inference



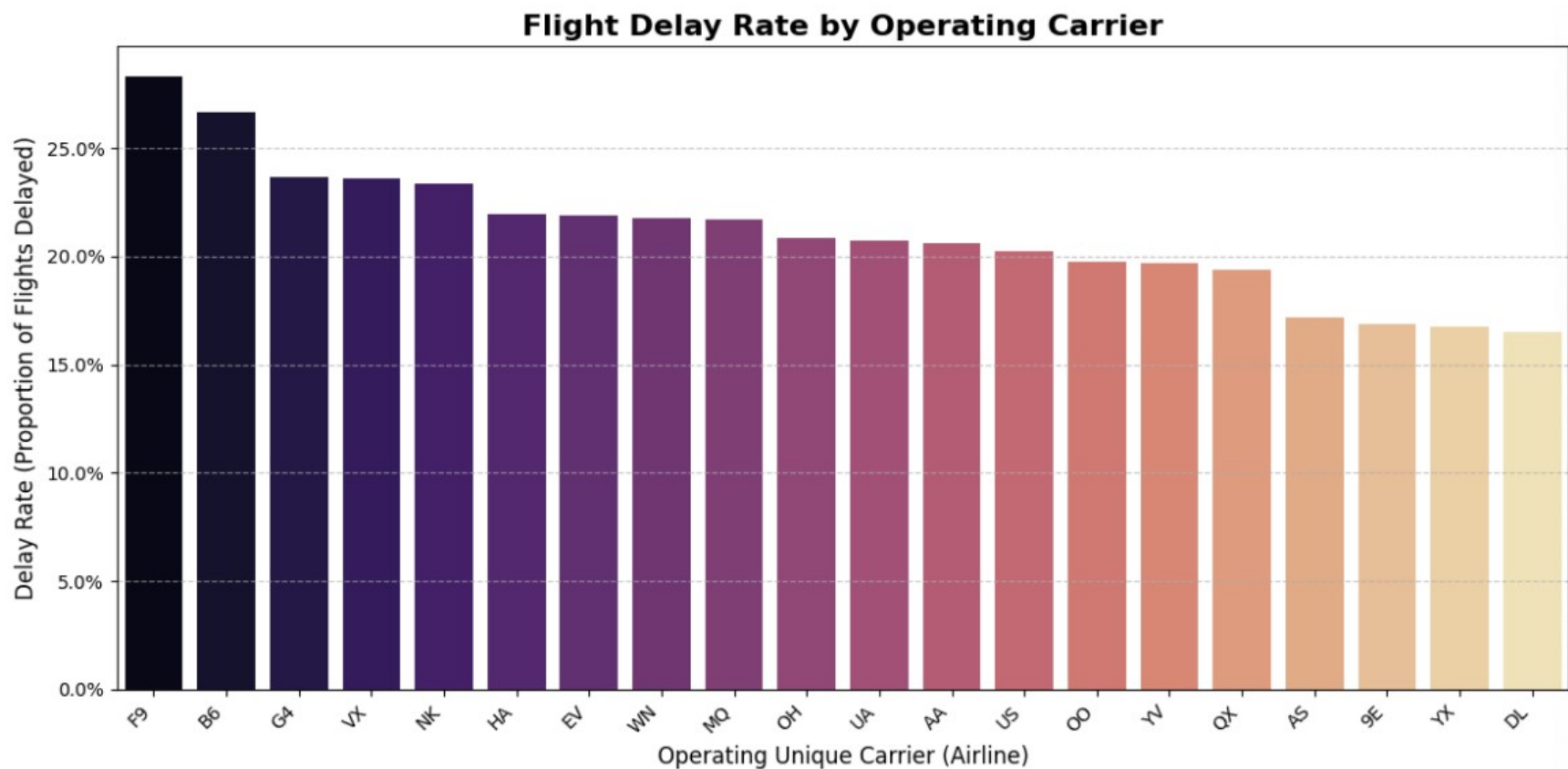
Model Comparison

	XGBoost	Random Forest
Accuracy	0.65	0.80
F-1 Score	0.69	0.71
Precision	0.76	0.64
Recall	0.65	0.80

Winter Holiday on Delays



Different Airlines on Delays



--- Carriers ranked by Delay Rate ---

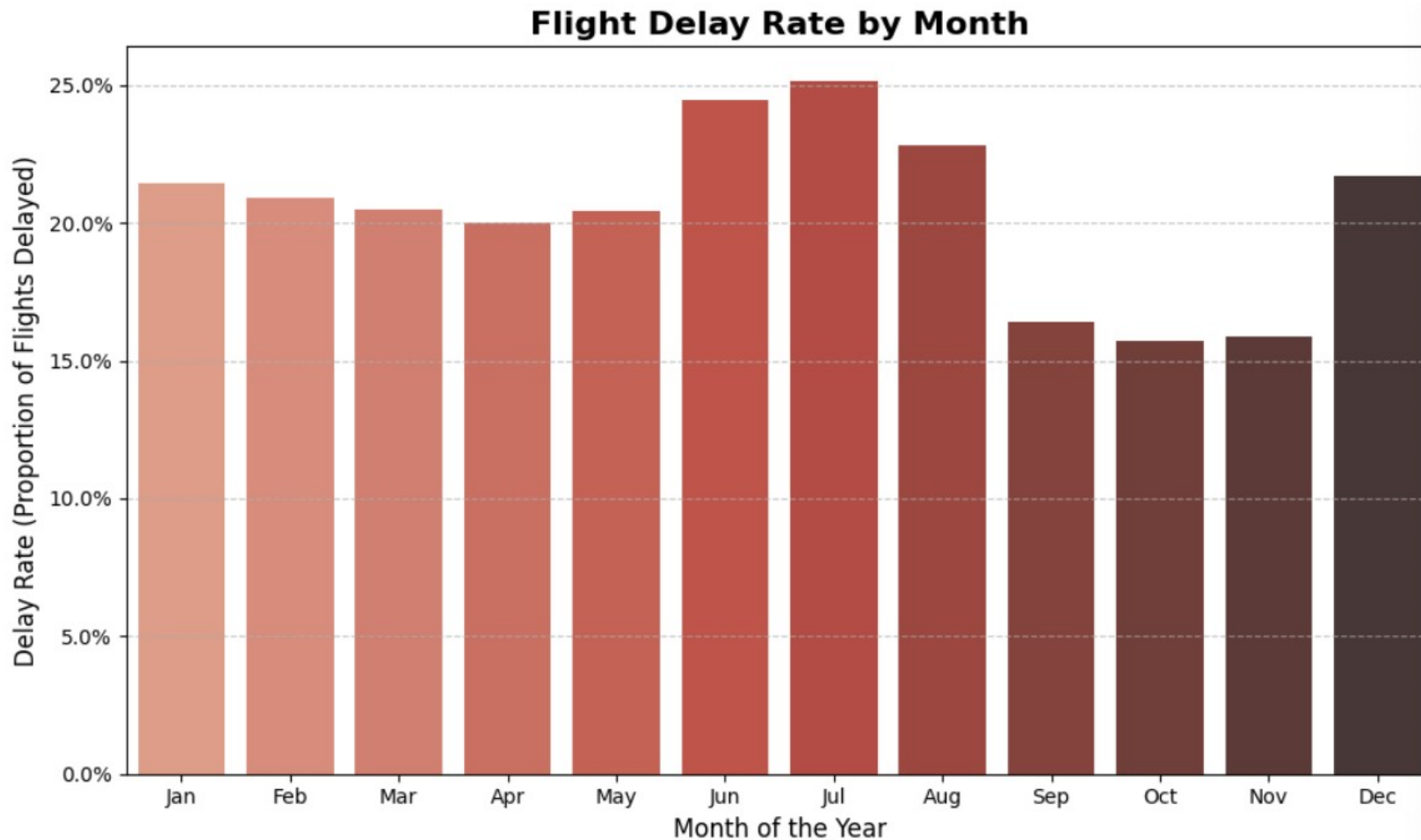
Top 5 Carriers:

Carrier	Delay_Rate
F9	0.2829
B6	0.2664
G4	0.2367
VX	0.2358
NK	0.2334

Bottom 5 Carriers:

Carrier	Delay_Rate
QX	0.1936
AS	0.1716
9E	0.1690
YX	0.1675
DL	0.1653

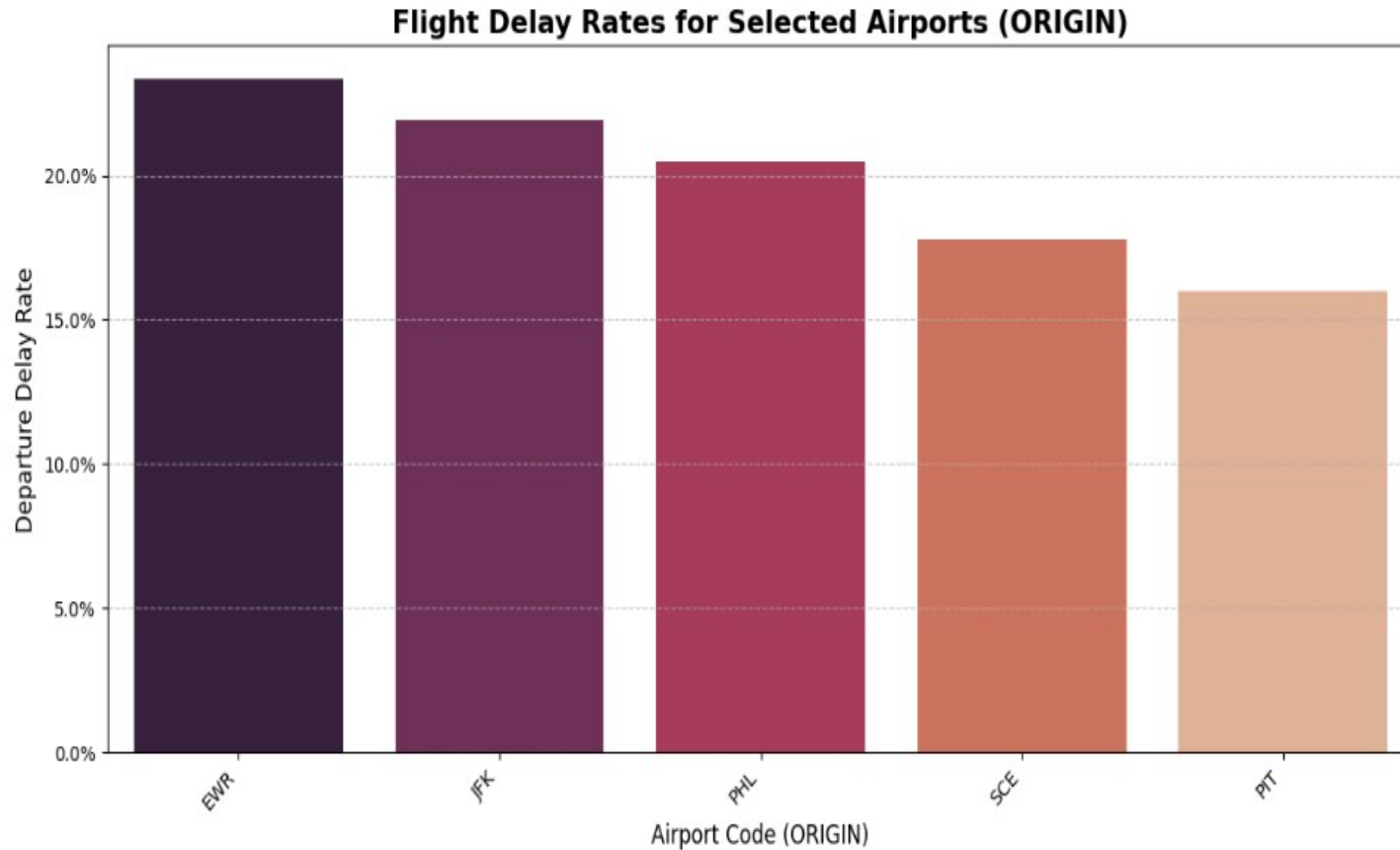
Different Months on Delays



--- Delay Rates by Month ---

Month_Name	Delay_Rate
Jan	0.2145
Feb	0.2089
Mar	0.2047
Apr	0.2003
May	0.2045
Jun	0.2447
Jul	0.2515
Aug	0.2282
Sep	0.1642
Oct	0.1572
Nov	0.1586
Dec	0.2173

Origin Airport on Delays



Flight Delay Rate for Specific Airports:

ORIGIN	Delay_Rate
EWR	0.2334
JFK	0.2191
PHL	0.2050
SCE	0.1776
PIT	0.1598

Thank you